



MATH140: Functions Cost Revenue Profit

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Evaluating cost and revenue functions

1. A firm's cost function is $C(x) = 15x + 112$, where x is units produced. Find $C(13)$ and interpret the result.

$$f(x) = 15x + 112, \quad x = 13$$

Answer: _____

2. If revenue is $R(x) = 22x + 36$ and cost is $C(x) = 14x + 336$, find profit $P(16) = R(16) - C(16)$.

$$f(x) = 22x + 36, \quad x = 16$$

Answer: _____

3. A company's revenue function is $R(x) = -3x^2 + 313$, where x is hundreds of units. Find $R(5)$.

$$-3x^2 + 313, \quad x = 5$$

Answer: _____

4. A firm's cost function is $C(x) = 12x + 304$, where x is units produced. Find $C(6)$ and interpret the result.

$$f(x) = 12x + 304, \quad x = 6$$

Answer: _____

5. If revenue is $R(x) = 42x + 8$ and cost is $C(x) = 23x + 383$, find profit $P(19) = R(19) - C(19)$.

$$f(x) = 42x + 8, \quad x = 19$$

Answer: _____

6. A company's revenue function is $R(x) = -5x^2 + 276$, where x is hundreds of units. Find $R(5)$.

$$-5x^2 + 276, \quad x = 5$$

Answer: _____



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Functions — Cost, Revenue & Profit Worksheet — MATH 140

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7. A firm's cost function is $C(x) = 39x + 538$, where x is units produced. Find $C(13)$ and interpret the result.

$$f(x) = 39x + 538, \quad x = 13$$

Answer: _____

8. If revenue is $R(x) = 35x + 64$ and cost is $C(x) = 27x + 182$, find profit $P(12) = R(12) - C(12)$.

$$f(x) = 35x + 64, \quad x = 12$$

Answer: _____

9. A company's revenue function is $R(x) = -1x^2 + 210$, where x is hundreds of units. Find $R(5)$.

$$-1x^2 + 210, \quad x = 5$$

Answer: _____

10. A firm's cost function is $C(x) = 16x + 324$, where x is units produced. Find $C(22)$ and interpret the result.

$$f(x) = 16x + 324, \quad x = 22$$

Answer: _____

11. If revenue is $R(x) = 25x + 25$ and cost is $C(x) = 14x + 254$, find profit $P(7) = R(7) - C(7)$.

$$f(x) = 25x + 25, \quad x = 7$$

Answer: _____

12. A company's revenue function is $R(x) = -5x^2 + 288$, where x is hundreds of units. Find $R(6)$.

$$-5x^2 + 288, \quad x = 6$$

Answer: _____

13. A firm's cost function is $C(x) = 45x + 200$, where x is units produced. Find $C(11)$ and interpret the result.

$$f(x) = 45x + 200, \quad x = 11$$

Answer: _____

14. If revenue is $R(x) = 34x + 10$ and cost is $C(x) = 30x + 335$, find profit $P(5) = R(5) - C(5)$.

$$f(x) = 34x + 10, \quad x = 5$$

Answer: _____



15. A company's revenue function is $R(x) = -2x^2 + 193$, where x is hundreds of units. Find $R(6)$.

$$-2x^2 + 193, \quad x = 6$$

Answer: _____

16. A firm's cost function is $C(x) = 44x + 188$, where x is units produced. Find $C(7)$ and interpret the result.

$$f(x) = 44x + 188, \quad x = 7$$

Answer: _____

17. If revenue is $R(x) = 48x + 56$ and cost is $C(x) = 19x + 229$, find profit $P(20) = R(20) - C(20)$.

$$f(x) = 48x + 56, \quad x = 20$$

Answer: _____

18. A company's revenue function is $R(x) = -2x^2 + 247$, where x is hundreds of units. Find $R(4)$.

$$-2x^2 + 247, \quad x = 4$$

Answer: _____

19. A firm's cost function is $C(x) = 12x + 259$, where x is units produced. Find $C(18)$ and interpret the result.

$$f(x) = 12x + 259, \quad x = 18$$

Answer: _____

20. If revenue is $R(x) = 24x + 14$ and cost is $C(x) = 23x + 339$, find profit $P(18) = R(18) - C(18)$.

$$f(x) = 24x + 14, \quad x = 18$$

Answer: _____

21. A company's revenue function is $R(x) = -5x^2 + 149$, where x is hundreds of units. Find $R(3)$.

$$-5x^2 + 149, \quad x = 3$$

Answer: _____

22. A firm's cost function is $C(x) = 15x + 250$, where x is units produced. Find $C(19)$ and interpret the result.

$$f(x) = 15x + 250, \quad x = 19$$

Answer: _____



23. If revenue is $R(x) = 40x + 65$ and cost is $C(x) = 29x + 104$, find profit $P(13) = R(13) - C(13)$.

$$f(x) = 40x + 65, \quad x = 13$$

Answer: _____

Domain of business functions

24. An average cost function is $AC(x) = 309 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{309}{x - 0}$$

Answer: _____

25. An average cost function is $AC(x) = 380 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{380}{x - 0}$$

Answer: _____

26. An average cost function is $AC(x) = 237 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{237}{x - 0}$$

Answer: _____

27. An average cost function is $AC(x) = 386 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{386}{x - 0}$$

Answer: _____

28. An average cost function is $AC(x) = 257 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{257}{x - 0}$$

Answer: _____



29. An average cost function is $AC(x) = 343 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{343}{x - 0}$$

Answer: _____

30. An average cost function is $AC(x) = 479 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{479}{x - 0}$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Evaluating cost and revenue functions, Domain of business functions. All answers verified by independent computation.

Solutions



Evaluating cost and revenue functions

1. A firm's cost function is $C(x) = 15x + 112$, where x is units produced. Find $C(13)$ and interpret the result.

$$f(x) = 15x + 112, \quad x = 13$$

$$\rightarrow C(13) = 15(13) + 112 = 195 + 112 = 307.$$

$\rightarrow$ The total cost to produce 13 units is \$307.

Answer: $f(13) = 307$

2. If revenue is $R(x) = 22x + 36$ and cost is $C(x) = 14x + 336$, find profit $P(16) = R(16) - C(16)$.

$$f(x) = 22x + 36, \quad x = 16$$

$$\rightarrow P(x) = R(x) - C(x) = (22 - 14)x + (36 - 336).$$

$$\rightarrow P(16) = 352 + 36 \text{ — you get the idea: compute } R(16) - C(16).$$

Answer: $f(16) = 388$

3. A company's revenue function is $R(x) = -3x^2 + 313$, where x is hundreds of units. Find $R(5)$.

$$-3x^2 + 313, \quad x = 5$$

$$\rightarrow R(5) = -3(5)^2 + 313 = -75 + 313 = 238.$$

Answer: $-3(5)^2 + 313 = -75 + 313 = 238$

4. A firm's cost function is $C(x) = 12x + 304$, where x is units produced. Find $C(6)$ and interpret the result.

$$f(x) = 12x + 304, \quad x = 6$$

$$\rightarrow C(6) = 12(6) + 304 = 72 + 304 = 376.$$

$\rightarrow$ The total cost to produce 6 units is \$376.

Answer: $f(6) = 376$

5. If revenue is $R(x) = 42x + 8$ and cost is $C(x) = 23x + 383$, find profit $P(19) = R(19) - C(19)$.

$$f(x) = 42x + 8, \quad x = 19$$

$$\rightarrow P(x) = R(x) - C(x) = (42 - 23)x + (8 - 383).$$

$$\rightarrow P(19) = 798 + 8 \text{ — you get the idea: compute } R(19) - C(19).$$

Answer: $f(19) = 806$

6. A company's revenue function is $R(x) = -5x^2 + 276$, where x is hundreds of units. Find $R(5)$.

$$-5x^2 + 276, \quad x = 5$$

$$\rightarrow R(5) = -5(5)^2 + 276 = -125 + 276 = 151.$$

Answer: $-5(5)^2 + 276 = -125 + 276 = 151$



7. A firm's cost function is $C(x) = 39x + 538$, where x is units produced. Find $C(13)$ and interpret the result.

$$f(x) = 39x + 538, \quad x = 13$$

$$\rightarrow C(13) = 39(13) + 538 = 507 + 538 = 1045.$$

$\rightarrow$ The total cost to produce 13 units is \$1045.

Answer: $f(13) = 1045$

8. If revenue is $R(x) = 35x + 64$ and cost is $C(x) = 27x + 182$, find profit $P(12) = R(12) - C(12)$.

$$f(x) = 35x + 64, \quad x = 12$$

$$\rightarrow P(x) = R(x) - C(x) = (35 - 27)x + (64 - 182).$$

$$\rightarrow P(12) = 420 + 64 \text{ — you get the idea: compute } R(12) - C(12).$$

Answer: $f(12) = 484$

9. A company's revenue function is $R(x) = -1x^2 + 210$, where x is hundreds of units. Find $R(5)$.

$$-1x^2 + 210, \quad x = 5$$

$$\rightarrow R(5) = -1(5)^2 + 210 = -25 + 210 = 185.$$

Answer: $-1(5)^2 + 210 = -25 + 210 = 185$

10. A firm's cost function is $C(x) = 16x + 324$, where x is units produced. Find $C(22)$ and interpret the result.

$$f(x) = 16x + 324, \quad x = 22$$

$$\rightarrow C(22) = 16(22) + 324 = 352 + 324 = 676.$$

$\rightarrow$ The total cost to produce 22 units is \$676.

Answer: $f(22) = 676$

11. If revenue is $R(x) = 25x + 25$ and cost is $C(x) = 14x + 254$, find profit $P(7) = R(7) - C(7)$.

$$f(x) = 25x + 25, \quad x = 7$$

$$\rightarrow P(x) = R(x) - C(x) = (25 - 14)x + (25 - 254).$$

$$\rightarrow P(7) = 175 + 25 \text{ — you get the idea: compute } R(7) - C(7).$$

Answer: $f(7) = 200$

12. A company's revenue function is $R(x) = -5x^2 + 288$, where x is hundreds of units. Find $R(6)$.

$$-5x^2 + 288, \quad x = 6$$

$$\rightarrow R(6) = -5(6)^2 + 288 = -180 + 288 = 108.$$

Answer: $-5(6)^2 + 288 = -180 + 288 = 108$

13. A firm's cost function is $C(x) = 45x + 200$, where x is units produced. Find $C(11)$ and interpret the result.

$$f(x) = 45x + 200, \quad x = 11$$

$$\rightarrow C(11) = 45(11) + 200 = 495 + 200 = 695.$$

$\rightarrow$ The total cost to produce 11 units is \$695.

Answer: $f(11) = 695$



14. If revenue is $R(x) = 34x + 10$ and cost is $C(x) = 30x + 335$, find profit $P(5) = R(5) - C(5)$.

$$f(x) = 34x + 10, \quad x = 5$$

$$\rightarrow P(x) = R(x) - C(x) = (34 - 30)x + (10 - 335).$$

$$\rightarrow P(5) = 170 + 10 \text{ — you get the idea: compute } R(5) - C(5).$$

Answer: $f(5) = 180$

15. A company's revenue function is $R(x) = -2x^2 + 193$, where x is hundreds of units. Find $R(6)$.

$$-2x^2 + 193, \quad x = 6$$

$$\rightarrow R(6) = -2(6)^2 + 193 = -72 + 193 = 121.$$

Answer: $-2(6)^2 + 193 = -72 + 193 = 121$

16. A firm's cost function is $C(x) = 44x + 188$, where x is units produced. Find $C(7)$ and interpret the result.

$$f(x) = 44x + 188, \quad x = 7$$

$$\rightarrow C(7) = 44(7) + 188 = 308 + 188 = 496.$$

$$\rightarrow \text{The total cost to produce 7 units is \$496.}$$

Answer: $f(7) = 496$

17. If revenue is $R(x) = 48x + 56$ and cost is $C(x) = 19x + 229$, find profit $P(20) = R(20) - C(20)$.

$$f(x) = 48x + 56, \quad x = 20$$

$$\rightarrow P(x) = R(x) - C(x) = (48 - 19)x + (56 - 229).$$

$$\rightarrow P(20) = 960 + 56 \text{ — you get the idea: compute } R(20) - C(20).$$

Answer: $f(20) = 1016$

18. A company's revenue function is $R(x) = -2x^2 + 247$, where x is hundreds of units. Find $R(4)$.

$$-2x^2 + 247, \quad x = 4$$

$$\rightarrow R(4) = -2(4)^2 + 247 = -32 + 247 = 215.$$

Answer: $-2(4)^2 + 247 = -32 + 247 = 215$

19. A firm's cost function is $C(x) = 12x + 259$, where x is units produced. Find $C(18)$ and interpret the result.

$$f(x) = 12x + 259, \quad x = 18$$

$$\rightarrow C(18) = 12(18) + 259 = 216 + 259 = 475.$$

$$\rightarrow \text{The total cost to produce 18 units is \$475.}$$

Answer: $f(18) = 475$

20. If revenue is $R(x) = 24x + 14$ and cost is $C(x) = 23x + 339$, find profit $P(18) = R(18) - C(18)$.

$$f(x) = 24x + 14, \quad x = 18$$

$$\rightarrow P(x) = R(x) - C(x) = (24 - 23)x + (14 - 339).$$

$$\rightarrow P(18) = 432 + 14 \text{ — you get the idea: compute } R(18) - C(18).$$

Answer: $f(18) = 446$



21. A company's revenue function is $R(x) = -5x^2 + 149$, where x is hundreds of units. Find $R(3)$.

$$-5x^2 + 149, \quad x = 3$$

$$\rightarrow R(3) = -5(3)^2 + 149 = -45 + 149 = 104.$$

Answer: $-5(3)^2 + 149 = -45 + 149 = 104$

22. A firm's cost function is $C(x) = 15x + 250$, where x is units produced. Find $C(19)$ and interpret the result.

$$f(x) = 15x + 250, \quad x = 19$$

$$\rightarrow C(19) = 15(19) + 250 = 285 + 250 = 535.$$

$\rightarrow$ The total cost to produce 19 units is \$535.

Answer: $f(19) = 535$

23. If revenue is $R(x) = 40x + 65$ and cost is $C(x) = 29x + 104$, find profit $P(13) = R(13) - C(13)$.

$$f(x) = 40x + 65, \quad x = 13$$

$$\rightarrow P(x) = R(x) - C(x) = (40 - 29)x + (65 - 104).$$

$$\rightarrow P(13) = 520 + 65 \text{ — you get the idea: compute } R(13) - C(13).$$

Answer: $f(13) = 585$



Domain of business functions

24. An average cost function is $AC(x) = 309 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{309}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

25. An average cost function is $AC(x) = 380 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{380}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

26. An average cost function is $AC(x) = 237 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{237}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

27. An average cost function is $AC(x) = 386 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{386}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

28. An average cost function is $AC(x) = 257 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{257}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

29. An average cost function is $AC(x) = 343 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{343}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$



30. An average cost function is $AC(x) = 479 / (x - 0)$. Find the domain and explain why $x = 0$ must be excluded.

$$g(x) = \frac{479}{x - 0}$$

→ The average cost is undefined when $x = 0$ (denominator = 0).

→ Domain: all $x > 0$, $x \neq 0$. In context, x must also be positive (can't produce negative units).

Answer: $x \neq 0$

